SMILENets: Audio Representation Learning via Neural Knowledge Distillation of Traditional Audio-Feature Extractors

Mina A. Nessiem*†, Mostafa M. Amin*†, Björn W. Schuller*‡
*Chair of Embedded Intelligence for Health Care and Wellbeing, University of Augsburg, Germany
†AI R&D Team, SyncPilot GmbH, Augsburg, Germany
‡GLAM – Group on Language, Audio, & Music, Imperial College London, UK
mina.nessiem@syncpilot.com, mostafa.amin@syncpilot.com, schuller@IEEE.com

2023 8th International Conference on Frontiers of Signal Processing (ICFSP) | 979-8-3503-0879-2/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICFSP59764.2023.10372936

***Abstract*—Considerable work has been invested over the years into crafting features for different modalities, to facilitate the modelling of information. Said work – and the knowledge it is based on – has thus far been unused as a prior in Deep Neural Networks (DNNs) being trained to learn representations of different modalities. The representations that DNNs have been extracting thus-far are only based on the data at hand, which may either not necessarily be the best representations possible, or depend on a pre-training the DNNs using data with a combined length of thousands to tens of thousands of hours. In this paper, we introduce SMILENets, DNNs that are trained as students to distil the feature knowledge of audio features such as the ComParE challenge features as extracted by the teacher openSMILE toolkit. We explore SMILENets in terms of their ability to extract said audio features for different datasets under in- and out-of training distribution conditions and in terms of their utility as neural feature extractors for neural networks that would model datasets under the same validation conditions. We show that SMILENets learn to distil, and thus, extract features accurately for in-distribution conditions, while exhibiting close results for out-of-distributions scenarios where the SMILENet has been trained on a representative training dataset.**

Index Terms: supervised learning, representation learning, knowledge distillation, audio feature extraction, computational paralinguistics

I. Introduction

In order for Deep Neural Networks (DNNs) to be able to model (i.e. classification or regression) input data such as audio, images, or video, the networks need to operate on the level of a lower-dimensional representation of the input that nonetheless captures the salient features of the raw input data. Since this paper focuses specifically on the audio input modality, it will use terms focused on audio and its' low-level features. The necessity of building a *representation* of the input audio places practitioners wishing to make use of the capabilities of DNNs at a crossroads; the practitioners either feed their networks with inputs that consist of tried-and-true hand-crafted features such as spectrograms, Mel-Frequency Cepstral Coefficients (MFCC), formants, vocal chord perturbation, and others [1] that are grounded in years of research into human speech and its characteristics, or they give the networks as input the raw data directly. The former depends on pre-existing programs and means that the task cannot be performed end-to-end using DNNs, while the latter forces the networks to learn to represent the data in lower dimensional representations that nonetheless capture the salient features of the raw data, a task called *representation learning* [2]. To learn a suitable representation, researchers frequently use audio datasets with sizes in the thousands [3] or tens- [4] or hundreds- [5] of thousands of hours to pre-train their networks, a task that requires large amounts of compute, and is not suitable for practitioners training on datasets several orders of magnitude smaller.

Traditional audio features (also known as audio parameters, used interchangeably), whose fundamental units are *Low-Level Descriptors* [1, pp. 179 – 180], "are extracted at approximately 100 frames per second with typical window sizes of 10-30 ms" [1, p. 179]. A selection of a combination of low-level descriptors (LLDs) is called a *feature set*. The choice of which feature set a model uses has a large impact on the performance of the model.

Feature sets have so far been human-crafted, either as brute-forced sets of features, such as the COMputational PARalinguistics challengeE (ComParE) 2013 feature set [6], which has 6 373 features or as minimalistic feature sets, such as the Geneva Minimalistic Acoustic Parameter Set (GeMAPS), comprising of 62 features, or the extended GeMAPS, which summarises 88 [7]. In order to calculate the feature values of a feature set, feature extractors have been and are used. Audio feature extractors, such as openSMILE [8] [9], librosa [10] or pyAudioAnalysis [11] are examples of actual programs that perform the calculations that produce the features defined by feature sets. A survey of notable baseline feature sets and their component features is found in [12].

DNNs with at least one hidden layer that uses a non-linearity of some sort have been proven to be *universal function approximators* [13]. A feature extractor that calculates the features of a fixed feature set can be considered as a multi-dimensional function, which means that neural networks can – in theory – be used to approximate feature extractors. In this vein, we introduce SMILENets – neural networks that have been trained in a supervised manner to reproduce audio feature sets from raw audio such as openSMILE would. Our approach can be thought of as an offline, feature-based knowledge, teacher-student approach to Knowledge Distillation [14]. *Knowledge Distillation* is a technique wherein a smaller,

rized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:26:01 UTC from IEEE Xplore. Restrictions a

Acronym	Ref.	Len. [hrs]	[#]
CPSD	[6]	1.05	2 542
SC^2	[6]	11.9	1 430
GEMEP	[6]	.867	1 260
SVC	[6]	8.43	2 763
DSD	[16]	2.78	1 555
SSC	[16]	1.17	911
URTIC	[17]	44.4	28 652
MPSSC	[17]	.347	828
CRIED	[18]	2.83	5 587
HSS	[18]	7.05	845
EmotAsS	[18]	9.17	10 627
USoMS	[18]	5.13	2 313
DLFD	[19]	4.6	13 409
SLEEP	[19]	17.7	16 462
STYRIALECTS	[19]	2.32	9 732
Σ		120	98 916
MASC	[20]	10.1	36 554
UCL-SBM	[20]	3.27	49

TABLE I: Summary statistics of the individual component datasets of the combined dataset used to train General-SMILENet, and the MASC and UCL-SBM datasets.

less-complex student model is trained based on the outputs of a larger, more-complex teacher model [15]. We argue that our approach can achieve similar results to traditional feature extractors for in-distribution datasets with just ten hours of audio, or with around 100 hours to also allow the modelling of out-of-distribution datasets. To the authors' knowledge, there has been no research in distilling traditional feature-sets despite this being an exciting research avenue that promises many concrete benefits.

This paper introduces our approach to distilling traditional feature extractors in Section II. Section III showcases the results of experiments using the distilled student networks. Finally, Section IV concludes the paper and puts forward suggestions for future work.

II. Approach

In this section, we introduce a type of knowledge-distilled, feature-extracting neural networks called SMILENets. In order to construct a SMILENet, audio features are extracted from an audio dataset using openSMILE. A SMILENet model is then trained to take the raw audio of the dataset as input and to produce audio features distilling those of the ComParE feature set as extracted via openSMILE as output. Section II-A introduces the background concepts necessary. Section II-B introduces the datasets selected to train and validate the SMILENets, while Section II-C introduces the architectures used to construct both the Mask-SMILENet and General-SMILENet introduced in this paper.

A. Background

The models trained in this work are all Convolutional Recurrent Neural Networks (CRNNs), which are composed of successive convolutional blocks (similar groupings of successive layers) followed by Long Short-Term Memory (LSTM) [21] recurrent layers, finishing with fully connected layers [22]. The purpose of the convolutional blocks is to successively extract more complex feature representations from their input signals, while the purpose of the LSTM layers is to examine the extracted features temporally. The LSTM features are passed on to fully connected layers that perform the final classification or regression tasks. The fully connected layers can either be time-distributed (as in they apply their transformation separately to each segment of the temporal sequence) or not (not considering a temporal dimension. Some of these layers use Dropout [23]. The models are trained to reduce either the Mean-Squared Error (MSE) or the Concordance Correlation Coefficient (CCC) [24] if they are regression models, or binary cross-entropy if they are classification models. This reduction happens by employing numerical optimisation algorithms such as ADAM [25] or Adadelta [26].

B. Datasets

Our work makes use of 17 datasets in total, whose specific names, total lengths and file counts are listed in Table I, split as follows:

- 15 ComParE challenge datasets from [6] [16] [17] [18], and [19]. The audio files of these datasets are combined together to form one large training set of raw audio for General-SMILENet. We refer to this collectively as the combined dataset.
- ComParE 2020 [20] challenge datasets MASC and UCL-SBM, which are highlighted briefly below, while the reader is referred to the paper for all of their details.

The MASC dataset (referred to from now on as the Mask dataset for readability) consists of a total of 36 554 files worth of one second of speech, where each file has an associated binary classification label of whether the speaker is wearing a mask or not. The Mask dataset is split into 10 895 files for training, 14 647 for development, and 11 012 for testing. A listing of notable baseline scores and scores of this work is provided in Table III and discussed deeper in Section III.

The UCL-SBM (referred to from now on as the Breathing dataset for readibility) dataset consists of 49 files, each consisting of four minutes of speech, where each file has an associated label of a numerical sequence of 6 000 values. The Breathing dataset is split into 17 files for training, 16 for development, and 16 for testing.

We first explored the distillation task by extracting audio features from the Mask dataset, a procedure that was then conducted on a larger scale using the combined dataset, explained below. The files of the dataset are all transformed into 16 kHz, mono-channeled wave files. For each of the one second files of the Mask dataset, the ComParE feature set is calculated for 10 ms windows. The ComParE feature set consists of 65 suprasegmental features and their deltas (change in feature values since the last sample), forming a total of 130 features, such that each example consists of 16 000 raw audio samples and each label of 100 windows of 130 features, each. The created features are then z-score normalised on a feature basis, bringing the means and variances of the different features to similar regions, a step that considerably eases the network training process. Further discussion of the training processes is explored in Section II-C.

rized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:26:01 UTC from IEEE Xplore. Restrictions a

Input (16000, 1)
Convolutional Block L: 3, F: 128, P: 10
Convolutional Block L:3, F: 256, P: 4
Convolutional Block L: 3, F:512, P:4
LSTM U: 128, R: True
LSTM U: 128, R: True
LSTM U: 128, R: True
Time-Dist (FC) U: 130
Output (100, 130)

(a) Mask-SMILENet

Input (16000, 1)
Convolutional Block L: 2, F: 32, P: 2, D: 0.1
Convolutional Block L: 2, F: 64, P: 2, D: 0.1
Convolutional Block L: 2, F: 128, P: 2, D: 0.1
Convolutional Block L: 2, F: 128, P: 2, D: 0.1
Convolutional Block L: 2, F: 256, P: 2, D: 0.1
Convolutional Block L: 2, F: 256, P: 2, D: 0.1
LSTM U: 128, R: True, D: 0.3
Time Dist (FC) U: 130
Time Dist (FC) U: 130
Output (100, 130)

(b) General-SMILENet architecture

TABLE II: Architecture diagrams of the two SMILENets, Mask-SMILENet and General-SMILENet. For the convolutional blocks, L stands for the number of 1-D convolutional layers per block, F for their number of convolutional filters, P for the pooling size of the 1-D blocks' MaxPooling layer and D for the dropout rate. For the LSTM and Time Distributed Fully-Connected layers, U refers to the number of units.

For the more general distilled network General-SMILENet (introduced in Section II-C), a more varied understanding of different raw audio distributions and their ComParE features was necessary, which is the reason why the 15 ComParE datasets were merged together to form the combined dataset. For this, the feature extraction process was the same as for the Mask dataset, except that a sliding window of one second length and sliding step size of one second was used to create the dataset examples, in order to enforce the (16 000, (100, 130)) structure explained above. Dataset files that did not satisfy the requirement of 16 000 samples for the audio or 100 feature windows were not included in the General-SMILENet training examples.

C. SMILENets Construction and Training

Both SMILENets are constructed as CRNNs, whose architectures are shown in Table II. Their convolutional blocks consist of layers of 1-D convolutions followed by a single 1-D MaxPooling layer. The convolutional layers use Rectified Linear Unit (ReLU) activations, same padding and kernel sizes that follow the formula pooling size * 2 // 3 + 1. The MaxPooling layers use a stride length of 2. The fully-connected layers use linear activations. Both networks were trained using the ADAM optimiser to minimise the MSE. We report a minimum validation loss of 0.136 for the Mask validation set using the Mask-SMILENet and a minimum validation loss of 0.2579 using the General-SMILENet on the merged dataset.

Model	Mask UAR[%]	Breathing r
Best* baseline	64.2	.507
2nd best* baseline	63.6	.498
Best ComParE+SVM	62.6	.244
ComParE†	**65.4**	**.508**
Mask-SMILENet†	64.0	—
General-SMILENet†	61.4	.499

TABLE III: Development-set model result statistics for the ComParE competition baseline models against the ComParE functionals and Mask- & General-SMILENet. UAR is the Unweighted Average Recall. r is Pearson's correlation coefficient. *Best single-model (i. e., no model-fusion), non-fused (i. e., trained only on the train dataset) model performance. †+CRNN.

III. Experiments and Results

In this section, we explore the Mask-SMILENet and the General-SMILENet in two different ways. The first, explored in Section III-A, concerns their ability to have learnt to extract or imitate the features of both in-distribution datasets (where the dataset was part of the training regime) or out-of-distribution datasets (not part of the training regime). The second, explored in Section III-B, concerns their utility as feature extractors for downstream neural networks that would model datasets (once again explored in both – in-distribution and out-of-distribution – datasets).

A. SMILENet Feature Imitation Experiments

In order to measure whether a SMILENet has learned to distill the audio features with respect to a certain dataset, the audio files are passed to the network, which predicts the feature sequences. For these experiments, and for the sake of easing the visual analysis of the results, we truncate the delta portion of both the original features and the predictions, focusing on the 65 primary LLD features. For each of the predictions, the MSE and Pearson Correlation Coefficient p are calculated. These are aggregated together on a per-feature basis for the entire dataset. This procedure is repeated for both the Mask-SMILENet and the General-SMILENet using the Mask and Breathing datasets. The results of the procedures are shown visually for both metrics in Figure 1.

The heatmaps show an interesting phenomenon, namely that the Mask-SMILENet is the most capable of both SMILENets when it comes to predicting the features of the Mask dataset, which is in-distribution, with the Mask-SMILENet showing the best MSE and p values of all of the experiments. For the Breathing dataset however, the Mask-SMILENet is shown to be completely ineffective at extracting features, exhibiting negligible p values and considerably worse MSE values of the experiments.

The General-SMILENet on the other hand is shown to have learnt to predict features for both the Mask and the Breathing datasets in a much more general manner, learning to predict features for both datasets with metrics similar to (but not quite as good as) those of Mask-SMILENet predicting the Mask Dataset. General-SMILENet, however, seems to find it difficult to predict certain features more than others, such as those of

rized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:26:01 UTC from IEEE Xplore. Restrictions a

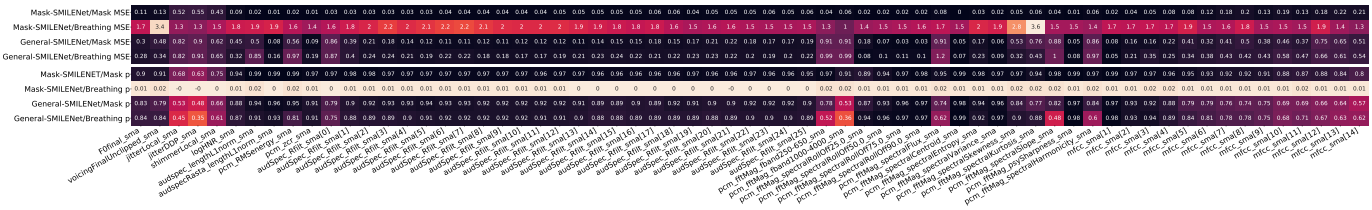


Fig. 1: Per-ComParE feature average MSE and Pearson correlation coefficients for the Mask and Breathing datasets as predicted by Mask- and General-SMILENet. Darker is better (lower MSE, higher Pearson)

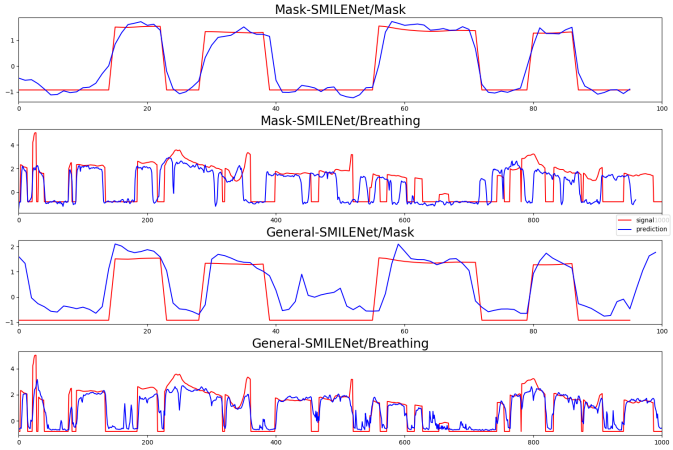


Fig. 2: Plots of the "F0final_sma" feature of the devel_00001 file of the Mask dataset and a 10 s subset of the devel_15 file of the Breathing dataset, plotted against the predictions of Mask-SMILENet and General-SMILENet of the same.

the following feature groups: jitter, shimmer, auditory spectral (audSpec), FFT bands, spectral slope, spectral harmonicity, MFCC.

The conclusions gleaned from the heatmaps can be confirmed visually when one looks at the feature plots of the predictions of the different model/dataset combinations for the "F0final_sma" (F_0 pitch) as provided in Figure 2.

B. SMILENet-based Dataset Modelling Experiments

To test the utility of SMILENets as feature extractors, it was necessary to see if they could be used to provide input features for neural network models to successfully model in-distribution and out-of-distribution datasets. In order to do so, we first use the ComParE features as extracted by openSMILE as inputs for CRNNs that model the Mask and Breathing datasets. We follow this by using the same CRNN architectures and hyperparameters to train CRNNs that instead use the outputs of each of the SMILENets (that are loaded as non-trainable weights) on the same datasets. The results of these experiments are shown, along with notable baseline scores in Table III. We first describe the architectures of each of the CRNNs used to model the datasets, followed by a discussion of the results.

The CRNN used to model the Mask dataset consists of a single convolutional block of a single 1-D convolution layer with 128 filters, a kernel size of three, a dropout rate of 0.3 and a ReLU activation. This is followed by a single 1-D MaxPooling layer with a pooling size of two. The convolutional block is followed by two LSTM layers of 16 units each, followed by a fully-connected layer of 32 units that each use ReLU activations, finishing with a fully-connected layer with a single, sigmoid activation neuron. The network is trained to minimize binary-crossentropy using the ADAM optimizer with a learning rate of 0.001 and a learning rate decay rate of $1\text{x}10^{-5}$.

For the Breathing dataset, the CRNN used for modelling consists of two convolutional blocks, four LSTM layers, followed by three time-distributed fully-connected layers, ending with a single, one-neuron fully-connected layer. Each of the convolutional blocks consists of three 1-D convolutional layers with a kernel size of seven and a dropout rate of 0.3, followed by a 1-D MaxPooling layer of pooling size two. The first convolution block uses (16, 32, 64) filters, while the second uses 128 filters for all of its convolutional layers. The LSTM layers all have 64 units and a dropout rate of 0.1, while the time-distributed fully-connected layers all have 128 units and a dropout rate of 0.1 as well. The network is trained to minimize the CCC using the Adadelta optimizer with a learning rate of 1 and a learning rate decay rate of $1\text{x}10^{-5}$.

As is shown in Table III, using the ComParE feature set as input to both modelling CRNNs allows the networks to exceed the best baseline metrics for both the Mask and the Breathing datasets, which showcases the usefulness of a powerful feature set. We then use the Mask-SMILENet to provide input features to the Mask-modelling CRNN, which produces results that are both close to the modelling based on the original ComParE features, as well as being better than the 2nd best baseline. Modelling both datasets using the General-SMILENet produces competitive results as well, showing that General-SMILENet does indeed learn to model out-of-distribution features, if to a degree of accuracy that is not as accurate as the Mask-SMILENet on in-distribution data, but nonetheless also better than the 2nd best baseline on the Breathing dataset. It should be noted that these results are not the best results the SMILENets are capable of providing, as experiments not shown within this work have shown that tuning the modelling CRNN for each of the SMILENets produces even better results, but the CRNNs were fixed for proper comparison of SMILENets.

IV. Conclusions and Future Work

In this paper, we introduced a class of feature-based knowledge, teacher-student Knowledge Distillation (KD) Deep Neural Networks (KNNs) we termed SMILENets. SMILENets distill audio feature sets using the audio feature extractor openSMILE. This was done by training two different Convolutional Recurrent

rized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:26:01 UTC from IEEE Xplore. Restrictions a

Neural Network (CRNN) SMILENets, each of which was trained using a different dataset, one using the Mask dataset and the other using a combined dataset of several others. For each of the datasets, their audio files were used to create pairs of training examples, of which the SMILENet training datasets were composed. The examples were each of one second raw audio segments with ComParE audio features extracted via SMILENet as labels.

By conducting analyses of the outputs of both networks given different in-distribution and out-of-distribution validation datasets, we showed that this approach is capable of learning to extract salient features from validation examples for in-distribution data and that it is capable of doing close to the same for out-of-distribution data if the network training set was of sufficient size and its examples come from different distributions.

A further set of experiments wherein different datasets were modelled using SMILENets as feature extractors were conducted. These experiments showed that if a SMILENet is used as a feature extractor for an in-distribution dataset, the dataset can be modelled to reach and exceed existing ComParE-based baselines. A SMILENet that is not trained on a large enough number of different distributions will be unable to extract meaningful features for an out-of-distribution dataset, rendering it useless for modelling. However, given a large enough training set composed of varied distributions, SMILENets can be used to model even out-of-distribution datasets to metrics near the top baseline values.

Given these encouraging results, the authors see a number of future avenues for research:

SMILENet-based Transfer Learning: This work opens the possibility of using pre-trained SMILENets as starting points for dataset modelling, rather than starting from random initializations. Audio networks have been trained from unlabeled videos [27], but not yet from existing feature sets.

Increase General-SMILENet Training Data and Accuracy: In this work, the General-SMILENet was trained using 120 hours of audio data and reached high levels of correlation with the normalised ComParE features of out-of distribution datasets. The authors are intrigued by the possibility of using much larger audio datasets to achieve in-distribution levels of accuracy for out-of-distribution datasets.

REFERENCES

[1] B. Schuller and A. Batliner, *Computational paralinguistics: emotion, affect and personality in speech and language processing*, first edition ed. Chichester, West Sussex ; Hoboken, N.J: John Wiley Sons, Ltd, 2014. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118706664

[2] Y. Bengio, A. Courville, and P. Vincent, "Representation learning: A review and new perspectives," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 35, no. 8, pp. 1798–1828, 2013.

[3] R. Zhang, H. Wu, W. Li, D. Jiang, W. Zou, and X. Li, "Transformer based unsupervised pre-training for acoustic representation learning," *arXiv preprint arXiv:2007.14602*, Jul. 2020.

[4] S. Chen, Y. Wu, C. Wang, Z. Chen, Z. Chen, S. Liu, J. Wu, Y. Qian, F. Wei, J. Li, and X. Yu, "UniSpeech-SAT: Universal Speech Representation Learning with Speaker Aware Pre-Training," *arXiv preprint arXiv:2110.05752*, Oct. 2021.

[5] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Advances in Neural Information Processing Systems*, H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, Eds., vol. 33. Curran Associates, Inc., 2020, pp. 12 449–12 460.

[6] B. Schuller, S. Steidl, A. Batliner, A. Vinciarelli, K. Scherer, F. Ringeval, M. Chetouani, F. Weninger, F. Eyben, E. Marchi, M. Mortillaro, H. Salamin, A. Polychroniou, F. Valente, and S. Kim, "The Interspeech 2013 Computational Paralinguistics challenge: Social Signals, Conflict, Emotion, Autism," in *Interspeech 2013*, pp. 148–152.

[7] F. Eyben, K. R. Scherer, B. W. Schuller, J. Sundberg, E. André, C. Busso, L. Y. Devillers, J. Epps, P. Laukka, S. S. Narayanan, and K. P. Truong, "The Geneva Minimalistic Acoustic Parameter Set (GeMAPS) for Voice Research and Affective Computing," *IEEE Transactions on Affective Computing*, vol. 7, no. 2, pp. 190–202, 2016.

[8] F. Eyben, M. Wöllmer, and B. Schuller, "Opensmile: The Munich Versatile and Fast Open-Source Audio Feature Extractor," in *Proceedings of the 18th ACM International Conference on Multimedia*, ser. MM '10. New York, NY, USA: Association for Computing Machinery, 2010, p. 1459–1462. [Online]. Available: https://doi.org/10.1145/1873951.1874246

[9] F. Eyben, F. Weninger, F. Gross, and B. Schuller, "Recent Developments in OpenSMILE, the Munich Open-Source Multimedia Feature Extractor," in *Proceedings of the 21st ACM International Conference on Multimedia*, ser. MM '13. New York, NY, USA: Association for Computing Machinery, 2013, p. 835–838. [Online]. Available: https://doi.org/10.1145/2502081.2502224

[10] B. McFee, C. Raffel, D. Liang, D. Ellis, M. McVicar, E. Battenberg, and O. Nieto, "librosa: Audio and music signal analysis in python," in *Proceedings of the 14th Python in Science Conference*. SciPy, 2015. [Online]. Available: https://doi.org/10.25080%2Fmajora-7b98e3ed-003

[11] T. Giannakopoulos, "pyaudioanalysis: An open-source python library for audio signal analysis," *PLOS ONE*, vol. 10, no. 12, pp. 1–17, 12 2015. [Online]. Available: https://doi.org/10.1371/journal.pone.0144610

[12] F. Eyben, *Real-Time Speech and Music Classification by Large Audio Feature Space Extraction*, 1st ed. Springer Publishing Company, 2016.

[13] K. Hornik, M. Stinchcombe, and H. White, "Multilayer feedforward networks are universal approximators," *Neural Networks*, vol. 2, no. 5, p. 359–366, Jul. 1989.

[14] J. Gou, B. Yu, S. J. Maybank, and D. Tao, "Knowledge distillation: A survey," *International Journal of Computer Vision*, vol. 129, no. 6, pp. 1789–1819, Jun 2021. [Online]. Available: https://doi.org/10.1007/s11263-021-01453-z

[15] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *arXiv preprint arXiv:1503.02531*, 2015.

[16] B. Schuller, S. Steidl, A. Batliner, J. Hirschberg, J. K. Burgoon, A. Baird, A. Elkins, Y. Zhang, E. Coutinho, and K. Evanini, "The INTERSPEECH 2016 Computational Paralinguistics Challenge: Deception, Sincerity & Native Language," in *Interspeech 2016*, pp. 2001–2005. [Online]. Available: http://dx.doi.org/10.21437/Interspeech.2016-129

[17] B. Schuller, S. Steidl, A. Batliner, E. Bergelson, J. Krajewski, C. Janott, A. Amatuni, M. Casillas, A. Seidl, M. Soderstrom, A. S. Warlaumont, G. Hidalgo, S. Schnieder, C. Heiser, W. Hohenhorst, M. Herzog, M. Schmitt, K. Qian, Y. Zhang, G. Trigeorgis, P. Tzirakis, and S. Zafeiriou, "The INTERSPEECH 2017 Computational Paralinguistics Challenge: Addressee, Cold & Snoring," in *Interspeech 2017*, pp. 3442–3446. [Online]. Available: http://dx.doi.org/10.21437/Interspeech.2017-43

[18] B. Schuller, S. Steidl, A. Batliner, P. B. Marschik, H. Baumeister, F. Dong, S. Hantke, F. B. Pokorny, E.-M. Rathner, K. D. Bartl-Pokorny, C. Einspieler, D. Zhang, A. Baird, S. Amiriparian, K. Qian, Z. Ren, M. Schmitt, P. Tzirakis, and S. Zafeiriou, "The INTERSPEECH 2018 Computational Paralinguistics Challenge: Atypical & Self-Assessed Affect, Crying & Heart Beats," in *Interspeech 2018*, pp. 122–126. [Online]. Available: http://dx.doi.org/10.21437/Interspeech.2018-51

[19] B. W. Schuller, A. Batliner, C. Bergler, F. B. Pokorny, J. Krajewski, M. Cychosz, R. Vollmann, S.-D. Roelen, S. Schnieder, E. Bergelson, A. Cristia, A. Seidl, A. S. Warlaumont, L. Yankowitz, E. Nöth, S. Amiriparian, S. Hantke, and M. Schmitt, "The INTERSPEECH 2019 Computational Paralinguistics Challenge: Styrian Dialects, Continuous Sleepiness, Baby Sounds & Orca Activity," in *Interspeech 2019*, pp. 2378–2382. [Online]. Available: http://dx.doi.org/10.21437/Interspeech.2019-1122

[20] B. W. Schuller, A. Batliner, C. Bergler, E.-M. Messner, A. Hamilton, S. Amiriparian, A. Baird, G. Rizos, M. Schmitt, L. Stappen, H. Baumeister, A. D. MacIntyre, and S. Hantke, "The INTERSPEECH 2020 Computational Paralinguistics Challenge: Elderly emotion, Breathing & Masks," in *Interspeech 2020*, Shanghai, China, September.

rized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:26:01 UTC from IEEE Xplore. Restrictions a

[21] S. Hochreiter and J. Schmidhuber, "LSTM Can Solve Hard Long Time Lag Problems," in *Proceedings of the 9th International Conference on Neural Information Processing Systems*, ser. NIPS'96. Cambridge, MA, USA: MIT Press, 1996, p. 473–479.

[22] E. Cakir, G. Parascandolo, T. Heittola, H. Huttunen, T. Virtanen, E. Cakir, G. Parascandolo, T. Heittola, H. Huttunen, and T. Virtanen, "Convolutional recurrent neural networks for polyphonic sound event detection," *IEEE/ACM Trans. Audio, Speech and Lang. Proc.*, vol. 25, no. 6, p. 1291–1303, jun 2017. [Online]. Available: https://doi.org/10.1109/TASLP.2017.2690575

[23] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, p. 1929–1958, Jan. 2014.

[24] I. Lawrence and K. Lin, "A concordance correlation coefficient to evaluate reproducibility," *Biometrics*, pp. 255–268, 1989.

[25] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *3rd International Conference on Learning Representations, ICLR 2015, San Diego, CA, USA, May 7-9, 2015, Conference Track Proceedings*, Y. Bengio and Y. LeCun, Eds., 2015. [Online]. Available: http://arxiv.org/abs/1412.6980

[26] M. D. Zeiler, "ADADELTA: An Adaptive Learning Rate Method," *ArXiv*, vol. abs/1212.5701, 2012.

[27] Y. Aytar, C. Vondrick, and A. Torralba, "Soundnet: Learning sound representations from unlabeled video," in *Proceedings of the 30th International Conference on Neural Information Processing Systems*, ser. NIPS'16. Red Hook, NY, USA: Curran Associates Inc., 2016, p. 892–900.

rized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:26:01 UTC from IEEE Xplore. Restrictions a